

GCE O-LEVEL MATHEMATICS PRACTICE PAPER 7

Exponentials & Logarithms

Duration: 2.5 hours

Total Marks: 100

Name: _____

Date: _____

SECTION A: Short Answer Questions (40 marks)

1. Solve: $2^x = 16$ [2 marks]

2. Solve: $\log_2(x) = 2$ [2 marks]

3. Solve: $3^x = 9$ [2 marks]

4. Find $\log_2(32)$ [2 marks]

5. Solve: $4^{x+1} = 64$ [3 marks]

6. Simplify: $\log(10) + \log(100)$ [2 marks]

7. Solve: $\log_2(x) + \log_2(2) = 1$ [3 marks]

8. If $\log_2(8) = x$, find x [2 marks]

SECTION B: Long Answer Questions (60 marks)

9. Solve: $2^{(2x-1)} = 8$. (4 marks)

10. Solve: $\log_{10}(x + 2) = 2$. (3 marks)

11. If $\ln(2) \approx 0.693$, $\ln(3) \approx 1.099$, find $\ln(12)$. (4 marks)

12. A population grows as $P = 1000 \times 2^{(t/5)}$, where t is time in years. Find population at $t = 10$. (3 marks)

13. Solve: $e^x = 10$ (use $\ln$). (3 marks)

14. Prove: $\log_a(x) + \log_a(y) = \log_a(xy)$. (4 marks)

SOLUTIONS

SECTION A SOLUTIONS

Q1: Solve $2^x = 16$

$$2^x = 2^4, x = 4$$

Q2: Solve $\log_3(x) = 2$

$$x = 3^2 = 9$$

Q3: Solve $3^x = 9$

$$3^x = 3^2, x = 2$$

Q4: Find $\log_2(32)$

$$2^5 = 32, \log_2(32) = 5$$

Q5: Solve $4^{(x+1)} = 64$

$$4^{(x+1)} = 4^3, x + 1 = 3, x = 2$$

Q6: Simplify $\log(10) + \log(100)$

$$= 1 + 2 = 3$$

Q7: Solve $\log_2(x) + \log_2(2) = 1$

$$\log_2(2x) = 1, 2x = 2, x = 1$$

Q8: Find $\log_2(8)$

$$2^3 = 8, x = 3$$

SECTION B SOLUTIONS

Q9: Solve $2^{(2x-1)} = 8$

$$2^{(2x-1)} = 2^3, 2x - 1 = 3, 2x = 4, x = 2$$

Q10: Solve $\log_3(x+2) = 2$

$$x + 2 = 3^2 = 9, x = 7$$

Q11: Find $\ln(12)$

$$\ln(12) = \ln(4 \times 3) = \ln(2^2) + \ln(3) = 2\ln(2) + \ln(3) = 2(0.693) + 1.099 = 2.485$$

Q12: Population at $t = 10$

$$P = 1000 \times 2^{(10/5)} = 1000 \times 2^2 = 1000 \times 4 = 4000$$

Q13: Solve $e^x = 10$

$$x = \ln(10) \approx 2.303$$

Q14: Prove $\log_a(x) + \log_a(y) = \log_a(xy)$

Let $\log_a(x) = p$, $\log_a(y) = q$, then $a^p = x$ and $a^q = y$, $xy = a^p \times a^q = a^{(p+q)}$, $\log_a(xy) = p + q = \log_a(x) + \log_a(y)$